

DIFFERENCE-AWARE RETRIEVAL POLICIES FOR IMITATION LEARNING

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ABSTRACT

Parametric imitation learning predicated on behavior cloning is notoriously susceptible to degraded generalization on out-of-distribution states, a pathology traceable to the compounding of prediction errors that accrues over the course of deployment. In the present work we demonstrate that the judicious reuse of the training corpus at inference time—operationalized through a semi-parametric, retrieval-based reformulation of imitation learning—materially attenuates this difficulty. We introduce **Difference-Aware Retrieval Policies for Imitation Learning (DARP)**, a semi-parametric retrieval-based approach that confronts this limitation by reparameterizing the imitation learning problem in terms of local neighborhood structure rather than a direct state-to-action mapping. In lieu of fitting a single global policy, DARP trains a model to predict actions from the k -nearest neighbors drawn from the expert demonstrations, together with their associated actions and the relative distance vectors separating neighbor states from query states. Crucially, DARP presupposes no assumptions beyond those already inherent to standard behavior cloning: it obviates the need for supplementary data collection, online expert feedback, and task-specific prior knowledge alike. Across a heterogeneous suite of domains encompassing continuous control and robotic manipulation, and across representations ranging up to high-dimensional visual features, we document consistent improvements of 15–46% relative to standard behavior cloning. Code and demonstrations will be released upon acceptance.

1 INTRODUCTION

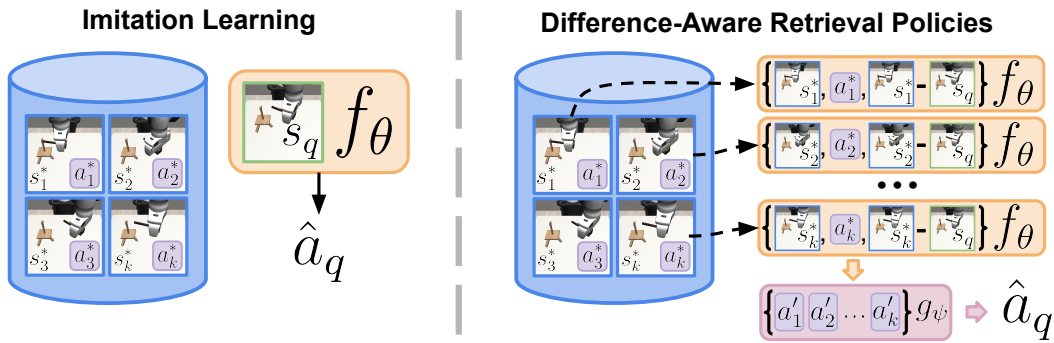


Figure 1: **Overview of DARP:** In contradistinction to standard BC (left), DARP (right) is organized around a retrieval-based reparameterization that hinges on the difference vectors relating query states to their retrieved neighbors. Whereas standard behavior cloning exploits the corpus of expert state-action pairs solely during training and then discards it at inference time, DARP retains this corpus and interrogates it via retrieval, thereby recovering a local neighborhood of expert state-action pairs surrounding each query point s_q .

Imitation learning through behavior cloning (BC) (Pomerleau, 1991) has furnished robots with the capacity to acquire complex, dexterous behaviors directly from expert demonstrations (Zhao et al., 2023; Chi et al., 2024; Black et al., 2024; Chung et al., 2014). Yet notwithstanding its conceptual and algorithmic simplicity, BC repeatedly proves brittle in practice, a fragility that is most pronounced

on long-horizon tasks (Ross et al., 2011). The root of the difficulty is *covariate shift*: infinitesimal errors, accumulating over the course of a rollout, progressively displace the agent into regions of the state space that the demonstration data represent only sparsely (Spencer et al., 2021; Ross et al., 2011). Within such out-of-distribution regions, BC policies become acutely unstable, engendering unreliable, high-variance behavior that culminates, more often than not, in outright failure.

This pathology is by now well documented, and a substantial body of work has sought to curtail compounding error (Ross et al., 2011; Venkatraman et al., 2015; Ke et al., 2024b; Levine et al., 2020). Such remedies, however, characteristically transcend the assumptions native to standard BC, presupposing simulators, interactive experts, copious quantities of sub-optimal data, or pronounced task-specific structure. Our ambition, by contrast, is to remain squarely within the pure BC regime—to learn exclusively from expert state–action pairs, unaided by any supplementary supervision or feedback. The pivotal question that thereby crystallizes is the following: *can the variance of BC policies be reduced using nothing more than the original demonstration dataset?*

From a statistical vantage point, BC minimizes the supervised risk on expert states alone. While this suffices to constrain bias on the training distribution, it leaves variance wholly unchecked: throughout the low-density regions of the state space—precisely those regions repeatedly traversed during closed-loop rollouts—the learned policy is at liberty to oscillate arbitrarily. A natural corrective is to impose *smoothness*, so that states in close proximity are mapped to commensurately similar predicted actions (Kobayashi, 2022; Asadi et al., 2018; Ke et al., 2024a; Chen et al., 2024); such a constraint suppresses spurious fluctuations and, in turn, confers greater rollout stability. A variety of mechanisms for eliciting smoothness have been investigated, which we survey in the related work of Section 4.

Although each is at times effective, each is likewise encumbered by characteristic shortcomings: augmentation furnishes no guarantee of consistency, global priors are prone to blurring behaviors that ought to remain distinct, temporal penalties exert their influence solely along the axis of time rather than that of space, and explicit graph regularizers demand the tuning of additional smoothness hyperparameters.

A complementary strand of research juxtaposes global and local paradigms of learning. “Global” supervised models (Black et al., 2024; Zhao et al., 2023; Chi et al., 2024) endeavor to compress the entirety of the demonstration dataset into a single parametric function, an approach that is characteristically brittle under distribution shift. “Local” methods (Pari et al., 2022; Mansimov & Cho, 2018; Salzberg & Aha, 1994), by contrast, tailor their predictions to the intrinsic structure of the dataset itself, consulting neighborhoods of kindred states and synthesizing outputs through non-parametric or semi-parametric operations upon the training distribution of expert behavior. This locality confers a measure of robustness, insofar as it forgoes dependence on any single parametric function; yet it is not without its own liabilities: its efficacy is inextricably contingent on the choice of distance metric, the naive averaging of a neighborhood is apt to blur otherwise distinct actions and to falter in representing multimodality, and the treatment of neighbors purely in terms of their absolute states can circumscribe generalization.

We introduce **Difference-Aware Retrieval Policies for Imitation Learning (DARP)**, a formulation that marries the robustness characteristic of local methods to the stability afforded by regularized global policy learning. At inference time, in place of predicting the action to be executed solely from the current query state via a feedforward pass through a parametric function, DARP (Fig. 1) first retrieves a set of k neighbors from the training corpus and subsequently predicts k candidate actions, each conditioned on a tuple comprising the neighbor state, its associated action, and its difference from the query state. These neighbor-informed predictions are thereafter aggregated in a permutation-invariant fashion to yield a single, robust action prediction. This design simultaneously grounds predictions in observed data—by virtue of non-parametric retrieval—and implicitly enforces local consistency—by virtue of parametric action prediction conditioned on the retrieved neighbors. We establish that this construction reduces variance while exacting no assumptions beyond those already demanded by standard Behavior Cloning: neither additional data, nor online supervision, nor task-specific knowledge is required. Cast in spectral terms, this mode of neighbor aggregation approximates a Laplacian filter on the k -NN (k -nearest neighbor) graph of expert states, thereby furnishing a parameter-free form of smoothing that adapts organically to the local density and geometry of the dataset.

We adduce both theoretical and empirical evidence that, while operating under precisely the same requirements as behavior cloning, DARP improves performance appreciably by curtailing variance and fortifying robustness to distribution shift. Our analysis formalizes the connection to Laplacian regularization, demonstrating that DARP implicitly applies a fixed low-pass spectral filter that attenuates high-frequency variance. Empirically, across our imitation learning evaluations, DARP realizes gains of 15–46% over conventional behavior cloning, spanning continuous control (MuJoCo), robotic manipulation (Robosuite, Robocasa), and high-dimensional visual imitation tasks (Robosuite with image state). We further demonstrate that DARP constitutes a general and scalable architecture that extends naturally to image-based domains and accommodates expressive policy classes such as transformers and Gaussian mixture models. Finally, we conduct a meticulous battery of ablations that isolates the contribution of our particular choice of representation and architecture, yielding general-purpose insights into retrieval-based algorithms for sequential decision-making problems.

2 DIFFERENCE-AWARE RETRIEVAL POLICIES FOR IMITATION LEARNING

In this work we instantiate a new class of imitation learning methods that appropriates the respective virtues of both “global” parametric approaches and “local” approaches. We advance a novel architecture, coupled with a deliberately simple training objective, that permits learning under exactly the requirements imposed by typical behavior cloning, while nonetheless furnishing substantial improvements both theoretically and empirically. Within this section we derive theoretical guarantees rigorously and from first principles. Readers whose interest lies chiefly in the practical experimental findings may proceed directly to Section 2.5 for a self-contained synopsis of DARP before turning to Section 3 for the empirical results.

By way of a preparatory foundation, we first delineate the problem setting (Section 2.1) and then examine a variant of regularized imitation learning (Section 2.2) that imposes additional structure drawn from the data, thereby securing improvements in variance, generalization, and stability. In Section 2.3, we proceed to demonstrate how the benefits of explicitly regularized learning may instead be realized implicitly, through a modification of the policy *architecture* rather than of the objective. Finally, in Section 2.4 we introduce our practical algorithm DARP, which actualizes these benefits by means of a semi-parametric, retrieval-augmented architecture that is broadly applicable to imitation learning with contemporary neural networks and generative modeling tools.

2.1 PRELIMINARIES: BEHAVIOR CLONING FOR IMITATION LEARNING

We situate our development within the canonical imitation learning setting, formalized as a finite-horizon Markov Decision Process (MDP), $\mathcal{M} = \{\mathcal{S}, \mathcal{A}, P_0\}$, in which $\mathcal{S}$ denotes the state space, $\mathcal{A}$ the action space, and P_0 the initial state distribution. A *policy* maps a state to a distribution over actions, $f_\theta : \mathcal{S} \rightarrow \Delta_{\mathcal{A}}$, in such a way as to maximize task-relevant objectives (for brevity, we suppress the subscript θ whenever discussing the general function class f). We presume access to expert, human-provided demonstrations $\mathcal{D}^*$, furnished as a collection of state-action pairs: $\mathcal{D}^* = \{(s_j^*, a_j^*)\}$. The notation s^* and a^* is reserved specifically to designate states and actions belonging to the expert dataset. The behavior cloning (Pomerleau, 1991) algorithm learns a policy f_θ from this dataset by recasting imitation as a conventional supervised learning problem — $\arg \max_\theta \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\log(f_\theta(a^* | s^*))]$. Although the distributional class of f_θ may in principle be an arbitrarily complex generative model (Lipman et al., 2023; Chi et al., 2024), we commence with a Gaussian parameterization in the interest of simplicity¹.

2.2 WARM-UP: NEIGHBOR MANIFOLD REGULARIZED IMITATION LEARNING

Whereas behavior cloning minimizes solely the supervised imitation loss over the expert states drawn from $\mathcal{D}^*$, such an objective in isolation exercises no control whatsoever over how the policy comports itself on states that depart from the manifold of expert states. In practice, accumulating errors propel the agent into out-of-distribution regions in which a BC policy may behave arbitrarily—a tendency especially acute for overparameterized neural networks (Ross & Bagnell, 2010).

¹We show that this can be relaxed in Section 2.4

To mitigate this deficiency, we observe that behavior cloning constrains function evaluations only at the training states, while leaving wholly unaccounted the relationships among states (and their corresponding actions) within a neighborhood, and thereby disregarding the underlying data manifold. In order to assimilate this information into policy learning, let us consider a modified objective that adjoins a regularization term explicitly promoting the *local consistency* of predictions: states lying in close proximity within the expert dataset ought to be mapped to correspondingly similar actions. This intuition gives rise to the following neighborhood-regularized loss ($\mathcal{L}_{\text{MRIL}}$), in which the standard imitation learning objective ($\mathcal{L}_{\text{BC}}$) is conjoined with an auxiliary smoothness penalty ($\mathcal{L}_S$) that compels predictions to honor the geometry of the dataset rather than to rely on pointwise supervision alone.

$$\mathcal{L}_{\text{MRIL}}(f) = \underbrace{\mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)]}_{\text{supervised risk}(\mathcal{L}_{\text{BC}})} + \lambda \underbrace{\mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right]}_{\text{smoothness regularizer}(\mathcal{L}_S)}, \quad (1)$$

where $\ell(f(s^*), a^*)$ denotes the supervised imitation loss, $\mathcal{N}_k(s^*)$ the set of k -nearest neighbors of s^* within the expert dataset, and the weights $w_i(s^*)$ the normalized kernel weights derived from the state differences — $w_i(s^*) \propto K_\Delta\left(\frac{\|s_i^* - s^*\|}{h}\right)$. As we discuss briefly below, and elaborate in full in Appendix A.1.1, this construction amounts to a form of *manifold regularization*, or Laplacian smoothing, whereby the policy incurs a penalty for high-frequency variation across the neighborhood of expert states. Such manifold regularization provably engenders improvements in policy variance, stability, and generalization alike.

Theorem 1 (Manifold Regularized BC ($\mathcal{L}_{\text{MRIL}}$) improves over vanilla BC ($\mathcal{L}_{\text{BC}}$)). *Let $f^* : \mathcal{S} \rightarrow \mathcal{A}$ denote the true, underlying expert policy, assumed to be C^2 -smooth on a compact state space $\mathcal{S}$, and let $f : \mathcal{S} \rightarrow \mathcal{A}$ denote the learned policy estimator. Consider the following two estimators, each trained on expert demonstrations:*

1. **Vanilla BC:** a global supervised model minimizing

$$\mathcal{L}_{\text{BC}}(f) = \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)].$$

2. **MRIL:** a neighbor-based estimator minimizing

$$\mathcal{L}_{\text{MRIL}}(f) = \mathcal{L}_{\text{BC}}(f) + \lambda \mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right],$$

where $w_i(s^*)$ are the kernel weights defined above and $\lambda > 0$.

Then, under the smoothness assumption on f , the following properties obtain:

- (i) Variance reduction: The Laplacian penalty in MRIL operates as a data-dependent Tikhonov regularizer, thereby yielding strictly smaller estimator variance than that of vanilla BC.
- (ii) Smoothness guarantee: Minimizers of $\mathcal{L}_{\text{MRIL}}$ satisfy a uniform bound on the local Lipschitz constant of f , whereas vanilla BC admits interpolants whose Lipschitz constants between training states may grow arbitrarily large.
- (iii) Policy stability: Over the course of a closed-loop rollout, the deviation recursion

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|^2$$

accumulates error linearly for vanilla BC, yet only sublinearly for MRIL, inasmuch as the smoothness regularizer enforces $\|f(s^*) - f(s'^*)\| = O(\|s^* - s'^*\|)$ for neighbors s^*, s'^* .

Taken together, these properties indicate that MRIL enjoys strictly superior generalization and stability guarantees relative to BC.

² L_s and L_a are the Lipschitz constants of the environment transition dynamics with respect to state and action, respectively.

Proof sketch. We relegate the detailed proof to Appendix A.1.1 and here confine ourselves to a brief sketch. The crux of the argument is to establish, first, that the smoothness regularizer corresponds precisely to a graph Laplacian penalty on the graph induced by a k -nearest neighbor (k -NN) affinity matrix defined through the kernel w_i . We then demonstrate that, as the number of samples tends to infinity, this graph Laplacian penalty converges to the weighted Dirichlet energy (Belkin & Niyogi, 2008; Zhou et al., 2003). Minimizing this Dirichlet energy accomplishes two ends simultaneously: (1) it guarantees that the learned f is locally Lipschitz almost everywhere, thereby securing smoothness and, in consequence, policy stability; and (2) it constitutes a form of Tikhonov regularization, thereby diminishing estimator variance while holding the bias in check. $\square$

Intuitively, the smoothness regularizer does not merely penalize pairwise disagreements among neighbors; rather, it drives the learned policy toward smoothness with respect to the underlying data manifold. Specifically, it contracts the local Lipschitz constant of f along those directions in which the data density $p(s)$ is high, thereby ensuring that infinitesimal changes in state induce correspondingly small and consistent changes in the predicted action. In consequence, the policy generalizes more dependably on in-distribution (ID) states and extrapolates in a structured, principled manner onto novel out-of-distribution (OOD) states residing in the neighborhood.

2.3 IMPLICIT MANIFOLD REGULARIZATION VIA IN-CONTEXT ARCHITECTURES

Although our MRIL objective does amortize local learning so as to deliver improvements over vanilla BC, it labors under two notable drawbacks. First, it introduces a hyperparameter λ whose value must be tuned to arbitrate between supervised accuracy and smoothness. Second, the imperative to optimize a modified, regularized objective in place of a standard BC objective may perturb the optimization landscape in deleterious ways. These considerations prompt a natural question: *can the very benefits conferred by MRIL (Eq. 1) be secured by modifying the policy architecture rather than the objective?*

In this section, we introduce a retrieval-based alteration of the policy architecture that induces an *implicit* manifold regularization effect (iMRIL), notwithstanding its reliance on a standard imitation objective. Through iMRIL, we recover the benefits of Laplacian smoothing (originally conferred by MRIL) while training on a standard BC objective (as illustrated in Fig. 2), and without introducing λ as an additional training hyperparameter. We subsequently build upon this algorithm to develop a practical instantiation of the method (DARP) in Section 2.4.

iMRIL architecture: The overarching idea underpinning iMRIL is disarmingly simple: we propose relocating the neighborhood aggregation (averaging) operation from the objective (as in Eq. 1) into the architecture itself. Thus, rather than learning a standard feedforward predictor $f(s)$ trained against a neighborhood-regularized smoothness objective (Eq. 1), we propose to embed the structure of neighborhood aggregation directly within the parameterization of the action predictor $\hat{f}$, all the while preserving the objective as one of standard imitation learning.

Concretely, iMRIL learns the parameters of a per-state predictor f_θ in such a manner that an action predictor explicitly parameterized via neighborhood aggregation, $\hat{f}(s^*) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s^*)} f_\theta(s_i^*)$, taken over the nearest-neighbor states of the training set $\{s_i^*\}_{i \in \mathcal{N}_k(s^*)}$, yields accurate predictions of the corresponding expert action a^* . Under this parameterization, iMRIL optimizes, at training time, the following:

$$\arg \min_{\theta} \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} \left[\left\| \underbrace{\left(\frac{1}{k} \sum_{i \in \mathcal{N}_k(s^*)} f_\theta(s_i^*) \right)}_{\hat{f}(s^*)} - a^* \right\|_2 \right] \quad (2)$$

At deployment time, inference on a new state s_q may be performed simply by retrieving the k -NN of s_q from the training set and carrying out neighborhood aggregation, $\hat{a} = \hat{f}(s_q) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} f_\theta(s_i^*)$.

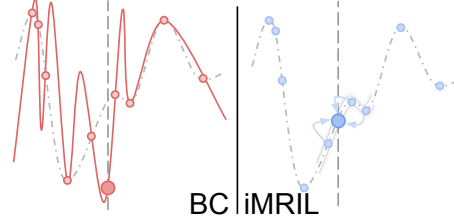


Figure 2: **iMRIL implicitly achieves Laplacian smoothing, thereby reducing variance and enforcing local consistency**, whereas the absence of any smoothness constraint in standard BC permits arbitrarily jagged function approximations.

As we demonstrate in Section 2.4, the particular parameterization of f is of decisive importance and exerts a pronounced influence on the empirical performance of iMRIL—a consideration that motivates the development of DARP.

Intuitively, we parameterize the action predictor $\hat{f}$ as an aggregation of the predictions $f(s_i^*)$ evaluated at neighbor states drawn from the training data, and then learn f . Supervising the post-aggregation function implicitly precludes any of the f predictions from being arbitrarily non-smooth, thereby conferring the benefits adduced in Section 2.2. We establish a direct equivalence between iMRIL and the Laplacian regularization of Section 2.2.

Equivalence between iMRIL and MRIL: While we relegate a full proof of the formal equivalence between MRIL and iMRIL to Appendix Section A.1, we state here our principal result together with a proof sketch to that effect.

Theorem 2 (iMRIL is parameter-free Laplacian regularization for BC (MRIL)). *Consider the symmetric normalized k -NN graph Laplacian L (defined in Section 2.2), with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$.*

The minimizers of, respectively, the explicit MRIL objective (Section 2.2) and the implicit iMRIL objective (Section 2.3) admit the following closed-form expansions:

$$f_{\text{MRIL}} = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j \quad \hat{f}_{\text{iMRIL}} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j$$

The neighbor aggregation step of iMRIL applies the fixed spectral filter $\phi_{\text{iMRIL}}(\mu) = 1 - \mu$ to the graph Laplacian L , thereby preserving low-frequency modes while suppressing high-frequency ones. The congruence between $\hat{f}_{\text{iMRIL}}$ and f_{MRIL} establishes that iMRIL is tantamount to a built-in form of Laplacian smoothing (MRIL) with an effective $\lambda \approx 1$ in normalized units. In contradistinction to explicit regularization, this implicit filter demands no additional hyperparameter tuning whatsoever.

Proof sketch. We relegate the full details to Appendix Section A.1.2. The explicit regularizer admits a spectral solution upon diagonalizing the k -NN Laplacian, which yields a filter of the form $(1 + \lambda \mu)^{-1}$ acting on each eigenmode. The implicit objective, for its part, may be recast as the neighbor aggregation $\hat{f} = S f$ with $S = D^{-1} A$ the random-walk matrix, which shares the same eigenvectors and applies the fixed filter $1 - \mu$. Intuitively, both operate as low-pass filters on the graph: modes associated with small eigenvalues (smooth variation across the data manifold) are largely preserved, whereas modes associated with large eigenvalues (rapid, high-variance fluctuations between neighbors) are strongly attenuated. Consequently, iMRIL implicitly performs Laplacian smoothing—reducing variance and enforcing local consistency—without any need to tune λ . $\square$

We stress that the implicit Laplacian smoothing perspective does not obviate the need to learn a policy; rather, it circumscribes the class of functions representable after aggregation. The neighbor-conditioned network f_θ learns how expert actions vary under local perturbations, proposing a locally adapted action for each neighbor, while the aggregation operator thereupon enforces variance reduction by smoothing these proposals across the neighborhood. In this manner, a clean division of labor emerges: learning supplies accuracy by correcting local bias, whereas aggregation supplies stability by controlling variance.

2.4 DIFFERENCE-AWARE RETRIEVAL POLICIES: A PRACTICAL INSTANTIATION OF iMRIL FOR IMITATION LEARNING

Building upon the conceptual framework of iMRIL, we now instantiate a practical algorithm suited to large-scale imitation learning. Taking the objective set out in Eq. 2 as our point of departure, we introduce a judicious choice of (1) parameterization and (2) neighbor aggregation, the conjunction of which yields strong empirical performance.

2.4.1 DIFFERENCE-BASED PARAMETERIZATION OF f_θ

The objective set forth in Eq. 2 leaves the parameterization and input representations of f_θ open to considerable latitude of interpretation. We observe that the neighborhood aggregation ought to learn how expert actions vary under local perturbations, which in turn suggests that f_θ should exploit knowledge of the *differences* between a query state and a neighbor state so as to adaptively propose a locally adapted action for each neighbor. In **Difference-Aware Retrieval Policies (DARP)**, accordingly, rather than parameterizing f_θ merely by $f_\theta(s_i^*)$, we furnish additional context in the form of the optimal neighbor action a_i^* together with the *difference* between the query state and the neighbor state, $\Delta s_i = s_i^* - s_q$; a predictor f_θ thereby predicts an action candidate a'_i for a query state s_q and a neighbor (s_i^*, a_i^*) by leveraging this difference information, as $a'_i = f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)$.

Let $\mathcal{N}_k(s_q)$ denote the index set of the k -nearest neighbors retrieved according to some distance function $d(s_q, s_i^*)$.³ To generate predictions with DARP, we may then perform the neighborhood aggregation delineated in Section 2.3, thereby predicting an action for any query state s_q :

$$\hat{a}_q = f_{\text{DARP}}(s_q) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} a'_i \quad (3)$$

$$= \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q). \quad (4)$$

At training time, this construction furnishes a straightforward imitation learning objective defined over the expert dataset $\mathcal{D}^*$:

$$\arg \min_{\theta} \mathbb{E}_{(s_q^*, a_q^*) \sim \mathcal{D}^*} [\|f_{\text{DARP}}(s_q^*) - a_q^*\|^2] \quad (5)$$

wherein we optimize the parameters of the predictor f_θ so as to minimize the discrepancy between the predicted action $\hat{a}_q$ and the optimal action a_q^* . Owing to the simplicity of the objective, any parameterization may serve for f_θ ; in our case, we adopt standard feedforward or convolutional neural networks. As we demonstrate in Section 3, this difference-based parameterization proves crucial to performance. At inference time, we generate the actions to be executed by retrieving the k -NN and performing inference through the neighborhood aggregation operation defined in Eq. 3.

2.4.2 GOING BEYOND LINEAR AGGREGATION

While the process of neighborhood aggregation has hitherto been restricted to a simple average over the neighborhood predictions, $\hat{a}_q = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} a'_i$, this is but a special case of a considerably broader class of permutation-invariant aggregation functions $g_\psi(\{a'_i\}_{i \in \mathcal{N}_k(s_q)})$. In particular, g_ψ could be parameterized by more expressive, set-compliant neural models such as the set transformer (Lee et al., 2019) or DeepSets (Zaheer et al., 2017). This observation intimates a generalization of the prediction model in Eq. 3, namely $\hat{a}_q = g_\psi(\{f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)\}_{i \in \mathcal{N}_k(s_q)})$. Beyond the attendant gains in expressivity, generalizing from a simple averaging operation to a parametric aggregation model g_ψ permits the representation of substantially richer action distributions (for instance, Gaussian mixture models (Pignat & Calinon, 2019) or diffusion models (Chi et al., 2024)) than the unimodal Gaussian implicit in the L_2 -regression objective of Eq. 5. In lieu of predicting $\hat{a}_q$ directly, DARP can instead predict the parameters α of an action distribution $p(a_q; \alpha)$ —the means, covariances, and mixing weights of a Gaussian mixture model, say, or the score function of a diffusion model. This latitude enables DARP to conduct maximum-likelihood training of multimodal action distributions rather than mere unimodal L_2 -regression:

$$\arg \max_{\theta} \mathbb{E}_{(s_q^*, a_q^*) \sim \mathcal{D}^*} [\log p(a_q^*; \alpha_\theta(s_q^*))], \text{ where } \alpha_\theta(s_q^*) = g_\psi(\{f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q^*)\}_{i \in \mathcal{N}_k(s_q^*)}) \quad (6)$$

Inference for a query state s_q proceeds by sampling $\hat{a}_q \sim p(\cdot; \alpha_\theta(s_q))$, where the parameters $\alpha_\theta(s_q) = g_\psi(\{f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)\}_{i \in \mathcal{N}_k(s_q)})$ are constructed from a set of neighbors retrieved at test time. We refer the reader to Appendix Section A.3 for detailed training pseudocode.

³In our work we employ the Euclidean distance within a pre-trained embedding space, although alternative neighborhood functions are equally admissible. We refer the reader to Appendix Section A.2.1 for a thorough discussion of the design decisions attending the construction of neighborhood sets via retrieval.

2.5 SUMMARY

We have shown that encouraging local consistency when fitting a behavior cloning policy is tantamount to manifold regularization (Laplacian smoothing) and provably improves variance, smoothness, and stability relative to standard BC (Theorem 1). Rather than adjoining an explicit regularization term—and its attendant hyperparameter λ —to our loss function, we have further shown that smoothing can be induced implicitly by embedding neighborhood aggregation within the policy architecture itself (Theorem 2). We are thus led to propose the following algorithm: at training time, for each expert state-action pair (s_q^*, a_q^*) , DARP retrieves the k -nearest neighbor states from $\mathcal{D}^*$, computes the difference vectors $\Delta s_i = s_i^* - s_q^*$ relating each neighbor to the query state, passes each neighbor tuple $(s_i^*, a_i^*, \Delta s_i)$ through a network f_θ to produce candidate actions, and finally aggregates these candidates by way of a permutation-invariant function g_ψ to predict the terminal action—all of which is trained exclusively with the standard imitation learning objective.

3 EXPERIMENTAL EVALUATION

We now turn to an empirical evaluation of DARP, organized around three pivotal questions: **Q1**: Can DARP consistently outperform standard behavior cloning?; **Q2**: Can DARP accommodate more complex state representations and action distributions?; and **Q3**: How do the individual architectural components contribute to DARP’s performance gains? To this end, we conduct experiments spanning multiple domains and employing low-dimensional state representations, high-dimensional image features, and a diverse assortment of action representations. Our evaluation encompasses continuous control tasks (MuJoCo), robotic manipulation (Robosuite), and purpose-built discontinuous environments that deliberately stress-test the neighbor-based approach.

3.1 BASELINE COMPARISONS AND TASK DESCRIPTIONS

MuJoCo Tasks: The MuJoCo (Todorov et al., 2012; Fu et al., 2020) tasks require controlling a variety of legged morphologies across multiple embodiments in order to effect forward locomotion over a flat plane. These tasks comprise Hopper (a single-legged hopping robot), Walker (a bipedal humanoid), Ant (a quadruped), and HalfCheetah (a biped).

Robosuite Tasks: The Robosuite (Zhu et al., 2020) tasks uniformly involve a single robotic arm manipulating objects. In the Stack task, the objective is to place a smaller cube atop a larger one; in the Threading task, to manipulate a thin, needle-like tool and insert it into a small ring; and in the Square Peg task, to manipulate a square wooden block, perforated at its center, and seat it upon a square peg.

RoboCasa Tasks: The RoboCasa (Nasiriany et al., 2024) tasks likewise center on a single robotic arm manipulating objects, here within a randomized kitchen setting. In the Drawer task, the objective is to close an open drawer; in the Door task, to close an open microwave oven door; and in the Stove task, to twist a knob so as to extinguish a stove burner.

Baseline Comparisons: We benchmark DARP against a spectrum of baselines and ablations: (1) **R&P** (Sridhar et al., 2025), which directly adopts the action associated with the nearest neighbor; (2) **LWR** (Pari et al., 2022), which performs locally weighted regression over the retrieved neighbors; (3) **BC**, standard parametric behavior cloning; (4) **REGENT** (Sridhar et al., 2025), a transformer-based in-context learning method conditioned on retrieved neighbors; and (5) **MRIL**, the explicitly smoothed variant of DARP delineated in Section 2.2.

3.2 CAN DARP CONSISTENTLY OUTPERFORM STANDARD BEHAVIOR CLONING? (Q1)

In this experiment, we interrogate DARP’s core hypothesis on tasks endowed with low-dimensional state representations, for which the inter-state distance metrics are well-defined and readily interpretable. The evaluation spans locomotion tasks drawn from MuJoCo together with robotic manipulation tasks drawn from Robosuite and RoboCasa, the latter furnished with data generated via MimicGen (Mandlekar et al., 2023). Throughout these experiments, the aggregation function g is realized as a simple average over all neighbor action predictions a' .

Method	Hopper	Ant	Walker	HalfCheetah
R&P (Sridhar et al., 2025)	711.82 $\pm$ 85.63	-305.97 $\pm$ 76.42	419.18 $\pm$ 50.21	-178.64 $\pm$ 29.75
LWR (Pari et al., 2022)	1703.78 $\pm$ 245.95	846.59 $\pm$ 216.06	1484.91 $\pm$ 356.54	1945.82 $\pm$ 567.26
BC	2313.65 $\pm$ 203.75	2376.20 $\pm$ 339.43	2658.40 $\pm$ 274.08	1063.23 $\pm$ 371.08
REGENT (Sridhar et al., 2025)	1819.39 $\pm$ 186.24	-302.10 $\pm$ 146.67	507.01 $\pm$ 76.10	169.85 $\pm$ 63.10
MRIL	2793.63 $\pm$ 156.41	3869.08 $\pm$ 241.00	4370.96 $\pm$ 168.13	701.58 $\pm$ 195.08
DARP	3545.57 $\pm$ 3.54	4383.28 $\pm$ 266.37	4894.01 $\pm$ 75.12	5515.41 $\pm$ 841.33
DARP Set Transformer	2965.86 $\pm$ 103.08	4063.79 $\pm$ 218.80	4752.42 $\pm$ 109.23	3417.85 $\pm$ 764.57

Table 1: **Both DARP and DARP Set Transformer surpass the competing approaches across all domains.** A performance comparison of DARP against BC and the remaining baselines across the MuJoCo environments under low-dimensional state. Reported scores are averaged over 100 independent trials, with 95% confidence intervals. The parametric policies employed Multi-Layer Perceptrons; for the corresponding results with diffusion policies, see Section 8.

We find that DARP delivers substantial improvements over standard behavior cloning across the entirety of the tested environments. Specifically, we observe performance gains on the order of 15–25 percentage points on the robotic manipulation tasks, alongside pronounced score improvements on the locomotion tasks (see Table 1 and Table 2). The purely non-parametric methods (R&P and LWR) fare poorly on these tasks, and although MRIL almost invariably attains a score exceeding that of vanilla BC, the highest scores across this suite are, without exception, achieved by our DARP architecture.

In light of the modifications introduced for the practical instantiation of Section 2.4, we next assess whether DARP scales to higher-dimensional input representations such as images.

3.3 CAN DARP HANDLE MORE COMPLEX STATE REPRESENTATION AND ACTION DISTRIBUTIONS? (Q2)

High-Dimensional Visual Input Representations. To probe the applicability of DARP beyond the regime of compact, low-dimensional states, we evaluate it on simulated robotic manipulation tasks employing R3M image

Method	Stack	Thrd.	Peg
BC	44	38	17
DARP	75	76	52

Table 3: Success rates (%) attained on the vision-based Robosuite tasks.



Figure 3: The “DrawerClose” RoboCasa task, in which the robotic arm is charged with closing the open drawer.



Figure 4: The “Threading” Robosuite task, in which the robotic arm is charged with inserting the needle implement into the small hole.

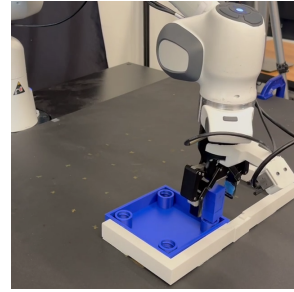


Figure 5: The real-world FurnitureBench square-table assembly task, in which the robot is charged with grasping a table leg and screwing it into a hole located at the corner of the table-top.

Method	RoboCasa			Robosuite			Real
	Drawer	Door	Stove	Stack	Thrd.	Peg	Sq. Table
BC	54	29	28	47	37	46	44
DARP	85	45	43	72	63	62	92

Table 2: A comparison of DARP against BC across all three evaluation environments under low-dimensional state features. Scores denote success percentages (RoboCasa & Robosuite: 100 trials; Real: 50 trials). DARP consistently outperforms BC across every task and environment, more than doubling the score of BC in the real world. The RoboCasa and Robosuite policies employed Multi-Layer Perceptrons, whereas the real-world results relied on a diffusion policy; see Section 8.

embeddings (Nair et al., 2022). This experiment adjudicates whether the neighbor-based approach retains its efficacy when states are represented as high-dimensional feature vectors extracted from visual observations (see Table 3). It is noteworthy that DARP not only outperforms standard BC, but that its average improvement of $\sim 35\%$ in fact exceeds the average improvement recorded on the Robosuite tasks under low-dimensional state ($\sim 22\%$). Empirically, this indicates that DARP adapted to complex, high-dimensional state representations more adeptly than did standard BC.

Multi-modal Action Distributions. We demonstrate that DARP is capable of solving complex multimodal imitation learning tasks, such as the Push-T environment, by a margin exceeding 20% over behavior cloning. Further details are relegated to Appendix A.2.2.

3.4 HOW DO DIFFERENT ARCHITECTURAL COMPONENTS CONTRIBUTE TO DARP’S PERFORMANCE GAINS? (Q3)

Ablation Study: In order to discern which components of the DARP architecture contribute most decisively to its performance gains, we undertake a comprehensive ablation study that scrutinizes each design choice in turn, namely: (1) standard DARP; (2) DARP divested of the neighbor actions; (3) an ensemble of 10 BC agents; (4) DARP with neighbors selected at random rather than by a distance metric; (5) DARP in which the L_2 norm of the distance vector is substituted for the vector itself; (6) the BC baseline, comprising the query state s_q alone; (7) DARP conditioned on the query state alone rather than on the difference vector between the query state and the neighbor states; and (8) DARP equipped with a permutation-dependent (that is, **not** permutation-invariant) aggregator for combining all of the a ’s. The results of this systematic ablation are reported in Figure 6.

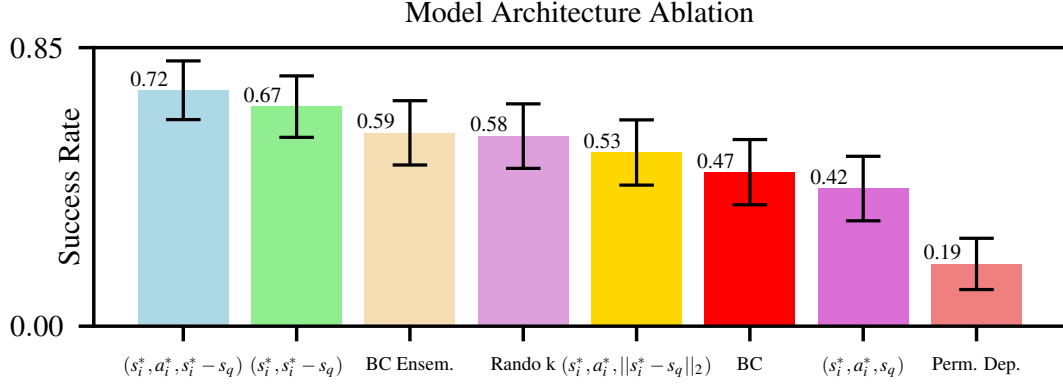


Figure 6: **Distance vectors and permutation invariance contribute preponderantly to DARP’s success.** An examination of how the performance of a DARP agent responds as successive modifications are applied to the core architecture reveals that its success is attributable, above all, to the distance vectors $(s_i^*, a_i^*, s_i^* - s_q)$. Success rate is averaged over 100 trials on the Robosuite Stack environment, with 95% confidence intervals.

The ablation study makes plain that distance vectors and permutation invariance are indispensable to DARP’s success, whereas the neighbor actions exert a comparatively modest influence. Random neighbor selection performs poorly, corroborating the claim that meaningful distance metrics governing neighbor selection are of the essence. The permutation-invariant aggregation function g likewise proves critical, insofar as permutation-dependent alternatives markedly degrade performance.

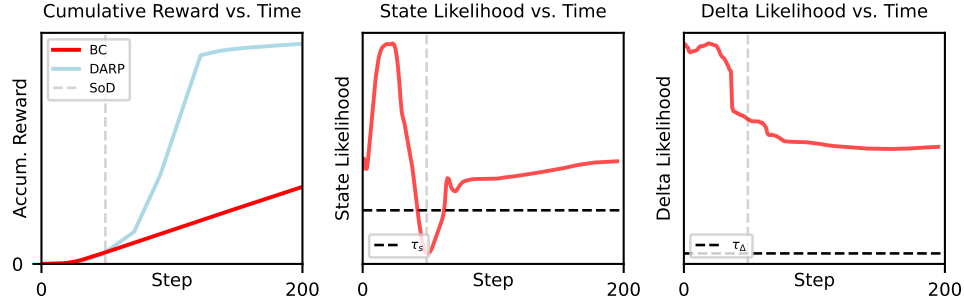


Figure 7: **The cumulative rewards accrued by BC and DARP on the Robosuite stack task exhibit initially identical rollouts that subsequently diverge, as BC fails the task while DARP succeeds.** A vertical dashed line marks the step at which the two diverge, labeled “SoD”. At the SoD, the state likelihood falls below τ_s (OOD), yet the delta likelihood remains above τ_Δ (in distribution).

Divergence Analysis: To elucidate more fully the provenance of DARP’s advantage over standard BC, we analyze the point of divergence within those rollouts in which the latter fails yet the former succeeds. We designate the “step of divergence” as the juncture at which DARP and BC begin to accrue significantly different reward. We define τ_s and τ_Δ as the 1st percentile of the training-set likelihoods (that is, 1% of the deltas encountered at training time are less probable than τ_Δ). Across all six rollouts spanning two distinct tasks (the Robosuite Stack task and the MuJoCo Hopper task), the query state at the SoD exhibits a state likelihood below τ_s yet a delta likelihood above τ_Δ . This finding lends further credence to our hypothesis that DARP’s gains are attributable, in part, to improved prediction on marginally out-of-distribution states, a consequence of reparameterizing in terms of the difference vectors to neighbors. (See Figure 7 for plots of the reward drift, the SoDs, and the state and delta likelihoods for one such task.) We refer the reader to Appendix A.2.4 for additional experiments concerning DARP’s robustness.

4 RELATED WORK

Non-Parametric Imitation Learning Methods: Non-parametric IL algorithms attain surprisingly strong performance by capitalizing on local structure. VINN Pari et al. (2022) investigates locally weighted regression for imitation, reporting surprising results within image embedding spaces, while MiDiGaP von Hartz et al. (2025) employs mixtures of Gaussian processes to model multimodal

trajectories and thereby achieve rapid generalization. SEABO Lyu et al. (2024) enlists retrieval methods in the service of offline RL, rewarding transitions in the vicinity of neighbors so as to constitute a reward function. FlowRetrieval Lin et al. (2024), STRAP Memmel et al. (2025), and Behavior Retrieval Du et al. (2023) conduct non-parametric retrieval and finetuning from large unlabeled datasets, effecting generalization through test-time training. DARP is distinguished from the foregoing by its idiosyncratic parameterization of retrieved states into $(s_i^*, a_i^*, s_i^* - s_q)$ tuples and by its learning of a semi-parametric policy, in contradistinction to reliance on purely non-parametric aggregation or test-time training. It is precisely this design that secures both the variance reduction characteristic of local methods and the generalization characteristic of parametric policies.

Smoothness-Constrained Policy Learning: A considerable body of recent literature has explored explicit smoothness constraints as a means of enhancing policy stability and robustness. L2C2 Kobayashi (2022) examines model-free RL under local Lipschitz continuity constraints, procuring smoothness and noise robustness without any forfeiture of expressiveness, while Asadi et al. (2018) advanced an analogous methodology for model-based RL under Lipschitz constraints. CCIL Ke et al. (2024a) extends these ideas to the generation of synthetic corrective labels for imitation learning by means of a Lipschitz-constrained dynamics model, an approach subsequently scaled up to humanoid controllers Chen et al. (2024) so as to attenuate shakiness upon deployment. DARP departs from these methods in that it enforces smoothness implicitly, through an architectural modification, all while retaining standard imitation learning objectives.

In-Context Learning Methods: Recent work has approached non-parametric retrieval from the vantage point of in-context imitation learning. REGENT Sridhar et al. (2025) investigates retrieval-augmented generalization by assimilating retrieved states, actions, and rewards into a causal transformer, whereas DPT Lee et al. (2023) harnesses supervised pretraining of transformers to predict actions from query states and in-context datasets, thereby effectively learning how to explore. Further in-context architectures include ICRT Fu et al. (2024), Instant Policy Vosylius & Johns (2025), and KAT Di Palo & Johns (2024). Whereas these methods aspire to rapid adaptation to novel tasks and environments, DARP is instead oriented toward securing higher performance and greater stability on standard imitation learning.

5 CONCLUSION

We have introduced **Difference-Aware Retrieval Policies for Imitation Learning (DARP)** (DARP), a nearest-neighbor-based algorithm that reparameterizes the imitation learning problem in terms of the relative differences between query states and their nearest neighbors, rather than in terms of a direct state-to-action mapping. We have proven that our method implicitly realizes Laplacian smoothing. Our experimental evaluation, conducted across a diverse array of domains encompassing continuous control and robotic manipulation, corroborates three central hypotheses. First, DARP consistently outperforms standard behavior cloning when operating on low-dimensional state representations. Second, DARP preserves its performance across disparate state representations, varying action-distribution modeling requirements, and heterogeneous task complexities, with improvements ranging from 15–46% across the scenarios examined. Third, our architectural ablations disclose that the distance vectors and permutation-invariant aggregation constitute the crucial ingredients of our algorithm.

6 REPRODUCIBILITY STATEMENT

Supplementary source code, the policy and environment configuration files, the MuJoCo experiment data, the scripts for generating expert demonstrations for the Robosuite tasks via MimicGen, and the modality transformation code will all be released upon acceptance. The results will be reproduced identically to those reported in the paper on NVIDIA L40 and L40s GPUs, with the sole exception of those results that necessitate a transformer (REGENT, Set Transformer), which, being non-deterministic, may deviate slightly from the reported figures.

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A APPENDIX

A.1 LEMMAS AND PROOFS

A.1.1 PROOF OF THEOREM 1

To establish Theorem 1, we begin by invoking a well-known result (Chung, 1997; Zhou et al., 2003; Belkin & Niyogi, 2008).

Lemma 1 (Smoothness regularizer as k -NN graph Laplacian penalty). *Let $\{s_1^*, \dots, s_n^*\}$ be the expert states with corresponding predicted actions $f(s_i^*) \in \mathbb{R}^{d_a}$. For each i , let $\mathcal{N}_k(s_i^*)$ denote the indices of the k -nearest neighbors of s_i^* (excluding i). Define asymmetric weights*

$$\tilde{W}_{ij} = \begin{cases} w_j(s_i^*), & \text{if } j \in \mathcal{N}_k(s_i^*), \\ 0, & \text{otherwise,} \end{cases}$$

and construct a symmetric affinity matrix

$$W_{ij} = \frac{1}{2}(\tilde{W}_{ij} + \tilde{W}_{ji}).$$

Let D be the degree matrix with $D_{ii} = \sum_j W_{ij}$, and define the k -NN graph Laplacian $L = D - W$.

Then the smoothness regularizer can be written as the quadratic form

$$\mathcal{L}_S(f) = \frac{1}{n} \sum_{i=1}^n \sum_{j \in \mathcal{N}_k(s_i^*)} w_j(s_i^*) \|f(s_i^*) - f(s_j^*)\|^2 \propto \text{Tr}(F^\top L F),$$

where $F = [f(s_1^*), f(s_2^*), \dots, f(s_n^*)]^\top \in \mathbb{R}^{n \times d_a}$. Equivalently, in the scalar case,

$$\mathcal{L}_S(f) \propto f^\top L f.$$

Corollary 1 (Continuum limit of smoothness regularizer). *Assume states $\{s_i^*\}_{i=1}^n$ are sampled i.i.d. from a smooth density $p(s^*)$ supported on an m -dimensional C^2 manifold $\mathcal{M} \subset \mathbb{R}^d$. Let W be the symmetrized k -NN affinity matrix constructed from a kernel K_Δ with bandwidth h , and let $L = D - W$ be the graph Laplacian.*

If $n \rightarrow \infty$, $h \rightarrow 0$, and $nh^{m+2} \rightarrow \infty$, then the normalized quadratic form converges to the weighted Dirichlet energy:

$$\frac{1}{n^2 h^{m+2}} \text{Tr}(F^\top L F) \longrightarrow C_K \int_{\mathcal{M}} \|\nabla_{\mathcal{M}} f(s)\|_2^2 p(s)^2 d\text{vol}(s),$$

where $C_K > 0$ is a constant depending only on the kernel K_Δ .

Theorem 1 (Manifold Regularized BC ($\mathcal{L}_{\text{MRIL}}$) improves over vanilla BC ($\mathcal{L}_{\text{BC}}$)). *Let $f^* : \mathcal{S} \rightarrow \mathcal{A}$ denote the true, underlying expert policy, assumed to be C^2 -smooth on a compact state space $\mathcal{S}$, and let $f : \mathcal{S} \rightarrow \mathcal{A}$ denote the learned policy estimator. Consider the following two estimators, each trained on expert demonstrations:*

1. **Vanilla BC:** a global supervised model minimizing

$$\mathcal{L}_{\text{BC}}(f) = \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)].$$

2. **MRIL:** a neighbor-based estimator minimizing

$$\mathcal{L}_{\text{MRIL}}(f) = \mathcal{L}_{\text{BC}}(f) + \lambda \mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right],$$

where $w_i(s^*)$ are the kernel weights defined above and $\lambda > 0$.

Then, under the smoothness assumption on f , the following properties obtain:

- (i) Variance reduction: The Laplacian penalty in MRIL operates as a data-dependent Tikhonov regularizer, thereby yielding strictly smaller estimator variance than that of vanilla BC.

(ii) Smoothness guarantee: Minimizers of $\mathcal{L}_{\text{MRIL}}$ satisfy a uniform bound on the local Lipschitz constant of f , whereas vanilla BC admits interpolants whose Lipschitz constants between training states may grow arbitrarily large.

(iii) Policy stability: Over the course of a closed-loop rollout, the deviation recursion

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|^4$$

accumulates error linearly for vanilla BC, yet only sublinearly for MRIL, inasmuch as the smoothness regularizer enforces $\|f(s^*) - f(s'^*)\| = O(\|s^* - s'^*\|)$ for neighbors s^*, s'^* .

Taken together, these properties indicate that MRIL enjoys strictly superior generalization and stability guarantees relative to BC.

Proof. We prove each claim in turn.

(i) Variance reduction. The Laplacian penalty in $\mathcal{L}_{\text{MRIL}}$ is

$$\sum_{i,j} W_{ij} \|f(s_i^*) - f(s_j^*)\|^2 = 2f^\top L f,$$

where $L = D - W$ denotes the graph Laplacian and W the k -NN affinity matrix. By Lemma 1, this quantity coincides with the empirical Dirichlet energy of f on the k -NN graph. It is well established (Zhou et al., 2003; Belkin & Niyogi, 2008) that such a quadratic penalty is equivalent to Tikhonov regularization with respect to the graph Laplacian norm $\|f\|_L^2 = f^\top L f$. Within statistical learning theory, the adjunction of a Tikhonov penalty strictly diminishes the variance of the estimator relative to the unregularized solution, all while holding the bias term in check. Consequently, MRIL enjoys smaller estimator variance than vanilla BC, which incorporates no such penalty.

(ii) Smoothness guarantee. Consider the continuum limit (Corollary 1): for i.i.d. samples $\{s_i^*\}$ from density p on a smooth manifold $\mathcal{M}$, the normalized penalty converges to

$$\int_{\mathcal{M}} \|\nabla f(s)\|^2 p(s)^2 d\text{vol}(s).$$

This quantity is the weighted Dirichlet energy of f on $\mathcal{M}$. Should this integral be finite, f resides in the Sobolev space $H^1(\mathcal{M}, p^2)$, and in particular f is locally Lipschitz almost everywhere, with

$$\|f(s^*) - f(s'^*)\| \leq C \|s^* - s'^*\| \quad \text{for } p\text{-a.e. neighbor pairs } s^*, s'^*.$$

Minimizers of $\mathcal{L}_{\text{MRIL}}$ therefore possess uniformly bounded local Lipschitz constants throughout the high-density regions of the state space. Minimizers of vanilla BC, by contrast, are subject to no such constraint: any oscillatory interpolant that fits the training data exactly incurs the same supervised risk, whence arbitrarily large Lipschitz constants become admissible.

(iii) Policy stability. Let $\Delta_t = \|s_t - s_t^*\|$ denote the deviation at time t . For Lipschitz dynamics T ,

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|.$$

Decompose the action error:

$$\|f(s_t) - f^*(s_t^*)\| \leq \|f(s_t) - f^*(s_t)\| + \|f^*(s_t) - f^*(s_t^*)\|.$$

For vanilla BC, the first term $\|f(s_t) - f^*(s_t)\|$ is minimized only on the empirical distribution $P_{\mathcal{S}}$; off-distribution, it may be $O(1)$ irrespective of Δ_t . The second term, by the smoothness of f^* , satisfies $\|f^*(s_t) - f^*(s_t^*)\| = O(\Delta_t)$. The recursion may thus assume the form

$$\Delta_{t+1} \leq L_s \Delta_t + L_a (O(1) + O(\Delta_t)),$$

which accumulates linearly in t .

⁴ L_s and L_a are the Lipschitz constants of the environment transition dynamics with respect to state and action, respectively.

For MRIL, the Laplacian penalty enforces

$$\|f(s^*) - f(s'^*)\| \leq C\|s^* - s'^*\| \quad \text{for neighbor pairs } (s^*, s'^*),$$

as shown in part (ii). Thus $\|f(s_t) - f^*(s_t)\| = O(h^2)$ (where h is the radius of the kernel bandwidth around the point s_t) by local-linear regression error bounds, and $\|f^*(s_t) - f^*(s_t^*)\| = O(\Delta_t)$ by smoothness of f^* . Combining these,

$$\Delta_{t+1} \leq L_s \Delta_t + L_a(O(h^2) + O(\Delta_t)).$$

Since the constant multiplying Δ_t is strictly smaller under the smoothness constraint, the cumulative error grows strictly more slowly than in vanilla BC. In particular, error growth is sublinear in the rollout horizon whenever h is small, in contrast to the linear growth exhibited by vanilla BC.

Conclusion. Claims (i)–(iii) jointly establish that MRIL affords lower variance, uniform smoothness control, and sublinear accumulation of rollout error relative to vanilla BC, thereby completing the proof. $\square$

Kernel choice. For the IC smoothness regularizer, we adopt a Gaussian kernel

$$w_i(s^*) \propto \exp\left(-\frac{\|s^* - s_i^*\|^2}{2h^2}\right), \quad \sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) = 1,$$

with bandwidth h set to the median distance to the k -th nearest neighbor across the dataset. This choice is standard within manifold regularization (Belkin & Niyogi, 2008; Zhou et al., 2003) and ensures the convergence of the graph Laplacian penalty to the Dirichlet energy in the continuum limit. In practice, we found this default to be stable across tasks, although other kernels (for instance, uniform k -NN or exponential decay) yield qualitatively comparable results.

A.1.2 PROOF OF THEOREM 2

We begin by establishing the requisite Lemmas, which characterize the spectral form of explicit Laplacian regularization and of neighbor aggregation, respectively.

Lemma 2 (Spectral form of explicit Laplacian regularization). *Let L be the symmetric normalized graph Laplacian with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$. The minimizer of the penalized objective*

$$\mathcal{L}_\lambda(f) = \|f - a^*\|^2 + \lambda f^\top L f$$

has the closed-form expansion

$$f_\lambda = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j.$$

Thus λ directly dictates the spectral filter $\phi_\lambda(\mu) = (1 + \lambda \mu)^{-1}$ applied to each Laplacian mode.

Proof. Diagonalize $L = U \Lambda U^\top$, with $U = [u_1, \dots, u_n]$ orthogonal and $\Lambda = \text{diag}(\mu_1, \dots, \mu_n)$, and write $f = U c$, $a^* = U b$ in this basis. The objective then becomes

$$\|U c - U b\|^2 + \lambda c^\top \Lambda c = \sum_{j=1}^n (c_j - b_j)^2 + \lambda \mu_j c_j^2.$$

Minimizing each term yields $c_j = \frac{1}{1 + \lambda \mu_j} b_j$, and transforming back into the original basis furnishes the stated expansion. $\square$

Lemma 3 (Spectral form of neighbor aggregation). *Let $S = D^{-1}A$ be the random-walk matrix of the k -NN graph, with adjacency A and degree D . For any prediction vector f , the neighbor-averaged prediction is $\hat{f} = S f$. In the Laplacian eigenbasis, this corresponds to the spectral filter*

$$\hat{f} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j,$$

i.e. $\phi_{\text{DARP}}(\mu) = 1 - \mu$.

Proof. By definition, $L = I - D^{-1/2}AD^{-1/2}$ and $S = D^{-1}A = I - L_{\text{rw}}$, where $L_{\text{rw}} = D^{-1}L$ denotes the random-walk Laplacian. Since L_{rw} and L share the same spectrum, up to a similarity transform, the eigenbasis $\{u_j\}$ diagonalizes S . Accordingly, for each mode u_j we have $Su_j = (1 - \mu_j)u_j$, which yields the claimed spectral filter. $\square$

Theorem 2 (iMRIL is parameter-free Laplacian regularization for BC (MRIL)). *Consider the symmetric normalized k -NN graph Laplacian L (defined in Section 2.2), with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$.*

The minimizers of, respectively, the explicit MRIL objective (Section 2.2) and the implicit iMRIL objective (Section 2.3) admit the following closed-form expansions:

$$f_{\text{MRIL}} = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j \quad \hat{f}_{\text{iMRIL}} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j$$

The neighbor aggregation step of iMRIL applies the fixed spectral filter $\phi_{\text{iMRIL}}(\mu) = 1 - \mu$ to the graph Laplacian L , thereby preserving low-frequency modes while suppressing high-frequency ones. The congruence between $\hat{f}_{\text{iMRIL}}$ and f_{MRIL} establishes that iMRIL is tantamount to a built-in form of Laplacian smoothing (MRIL) with an effective $\lambda \approx 1$ in normalized units. In contradistinction to explicit regularization, this implicit filter demands no additional hyperparameter tuning whatsoever.

Proof. From Lemma 3, the neighbor aggregation operator $S = D^{-1}A$ acts on Laplacian eigenmodes u_j as

$$Su_j = (1 - \mu_j)u_j,$$

where μ_j are the normalized Laplacian eigenvalues. Hence, in the graph Fourier basis, neighbor aggregation amounts to multiplying each mode by the fixed spectral filter $\phi_{\text{DARP}}(\mu) = 1 - \mu$.

Lemma 2, on the other hand, demonstrates that explicit Laplacian regularization with parameter λ yields the spectral filter $\phi_\lambda(\mu) = (1 + \lambda \mu)^{-1}$. Both filters downweight the high-frequency modes ($\mu \gg 0$) while preserving the low-frequency modes ($\mu \approx 0$); the salient distinction is that $\phi_\lambda(\mu)$ requires the tuning of λ , whereas $\phi_{\text{DARP}}(\mu)$ is parameter-free.

To see the equivalence, note that for small μ ,

$$\phi_{\text{DARP}}(\mu) = 1 - \mu \approx (1 + \mu)^{-1} = \phi_{\lambda=1}(\mu) \quad \text{up to } O(\mu^2) \text{ terms.}$$

Thus DARP may be construed as performing Laplacian smoothing with an effective regularization weight of order $\lambda \approx 1$ in normalized units. Moreover, for large μ , $\phi_{\text{DARP}}(\mu)$ damps the high-frequency modes yet more aggressively by driving them toward zero, thereby affording a sharper low-pass effect than explicit regularization.

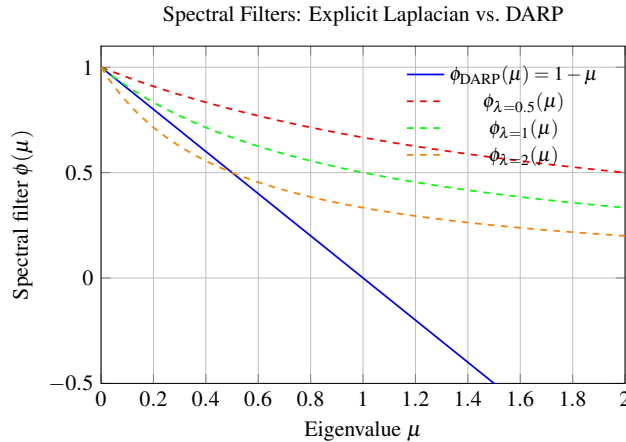


Figure 8: DARP effects sharper low-pass filtering.

We conclude, therefore, that DARP’s aggregation step is mathematically equivalent to implicit Laplacian regularization under the fixed spectral filter ϕ_{DARP} , obviating any need to tune λ explicitly. $\square$

DARP may therefore be regarded as a form of *locally adaptive implicit regularization*: in lieu of introducing an explicit global weight λ , its neighbor aggregation step enforces smoothness automatically by way of the graph structure. The effective regularization strength thus varies with the local degree and the neighborhood geometry, adapting itself to the density of the expert demonstrations. Spectrally, this behavior corresponds to the fixed filter $\phi_{\text{DARP}}(\mu) = 1 - \mu$, which suppresses the high-frequency modes more aggressively than any fixed explicit λ . Figure 8 renders this comparison graphically, exhibiting how DARP attains sharper low-pass filtering in the absence of any hyperparameter tuning.

A.2 ADDITIONAL EXPERIMENTAL DETAILS

A.2.1 RETRIEVAL

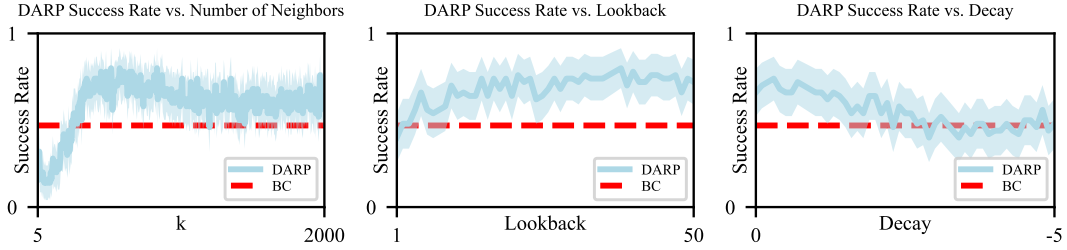


Figure 9: An analysis of DARP performance as the retrieval hyperparameters are swept: (left) the performance of a DARP model is poor when few neighbors are used, attains a global optimum upon retrieving approximately 500 neighbors, and plateaus just above BC’s success rate as k approaches the size of the dataset; (center) the performance of a DARP model generally improves, if only slightly, as more history is taken into account, and falls below that of BC solely when very little or no history is considered; (right) the performance of a DARP model is sensitive to the weight accorded to older observations during retrieval. Intuitively, if this decay is set too high, DARP performance becomes nearly indistinguishable from that obtained with little or no lookback, and thus falls below that of BC. Success rate is measured over 50 trials on the Robosuite Stack environment, with 95% confidence intervals included.

The selection of the distance function $d(s_q, s_i^*)$ that governs the choice of the k neighbors is of critical importance. While we find that a simple Euclidean distance between states can suffice, in our experiments we employ a slightly modified algorithm that exploits the fact that we are working with sequences of states and accordingly incorporates history into the distance calculation.

Suppose we are given a query trajectory $S_q = (\dots, s_{q,-1}, s_{q,0})$, in which $s_{q,0}$ denotes the current query state s_q , and suppose we wish to compute $d(s_q, s_i^*)$ for some state s_i^* drawn from the expert dataset. We first identify the trajectory from which this state originates—call it S_j^* —together with the index of s_i^* within that trajectory—call it t —so that s_i^* may be rewritten as $s_{j,t}^*$. Then, given a lookback parameter ℓ that specifies how many past states are to be considered, we obtain:

$$d(s_q, s_i^*) = \sum_{n=0}^{\ell-1} \|s_{q,-n} - s_{j,t-n}^*\|$$

This is simply the accumulation of the Euclidean distances between the current and preceding $\ell - 1$ states of the query trajectory and those of the source trajectory, presuming the indices to be valid. In practice, of course, we generally wish to accord greater emphasis to the more recent states, so that they exert a correspondingly greater influence upon the selection of neighbors. Accordingly, given some rate of exponential decay $\gamma \geq 0$, we have

$$d(s_q, s_i^*) = \sum_{n=0}^{\ell-1} \|s_{q,-n} - s_{j,t-n}^*\| \cdot e^{-\gamma n}$$

We refer the reader to Figure 9 for an experimental analysis of how the success rate within an environment varies as these parameters are swept.

A.2.2 CAN DARP HANDLE TASKS REQUIRING THE REPRESENTATION OF MULTI-MODAL ACTION DISTRIBUTIONS?

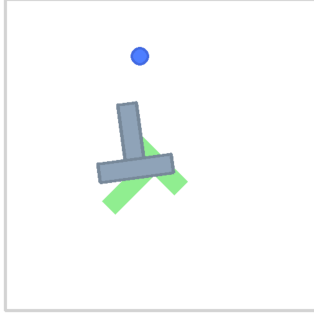


Figure 10: **Push-T Environment.** The objective is to control the blue circle so as to push the T-shaped block.

Method	Score
BC	48 ± 8
DARP	70 ± 8

Table 4: **Push-T Results.** Averaged over 100 trials, DARP surpasses BC.

We assess DARP’s capacity to accommodate complex action distributions by evaluating it on the Push-T task described in (Chi et al., 2024), which demands the representation of multi-modal action distributions (see Figure 10 for a visualization). For this experiment, DARP is furnished with a Set Transformer head that predicts the parameters of a Gaussian Mixture Model. We observe that DARP, so equipped with a GMM head, handles multi-modal distributions effectively, registering a 22% improvement over BC on the Push-T task (Q2) (see Table 4). This result demonstrates that DARP can be further adapted to the exigencies of multi-modal action-distribution modeling.

A.2.3 CAN DARP HANDLE DISCONTINUOUS ENVIRONMENTS WHERE NEARBY STATES MAY REQUIRE OPPOSING ACTIONS?

A salient concern for neighbor-based approaches is their performance in environments marked by strong discontinuities, wherein states proximate in Euclidean distance may nonetheless demand drastically divergent actions. To confront this concern, we devise a stress test predicated on a modified version of D4RL’s Umaze environment (see Figure 11 for a visualization).

Even within this deliberately adversarial, discontinuous environment, DARP attains a 57% success rate as against BC’s 25%. (Q3) (see Table 5) This outcome intimates that the distance vectors and permutation-invariant aggregation assist the model in discriminating between suitable and unsuitable neighbors, even where spatial proximity affords no guarantee of action similarity.

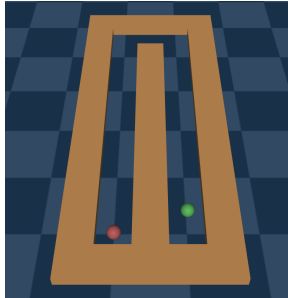


Figure 11: **Long maze environment.** The objective is to convey a force-actuated ball from the green starting location to the red destination.

Method	Succ. (%)
BC	25
DARP	57

Table 5: **Long maze results.** Averaged over 100 trials, DARP substantially outperforms BC.

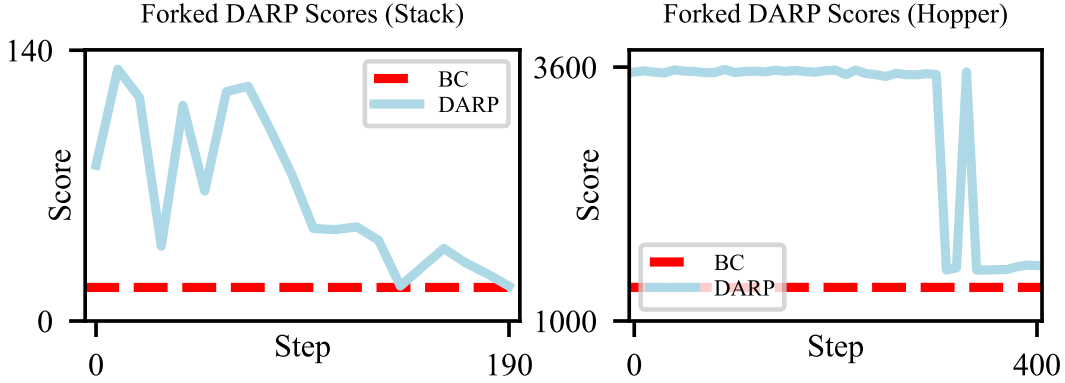


Figure 12: Across two distinct tasks (the Robosuite Stack task and the MuJoCo Hopper task), we roll out a BC agent and, at every k steps (here $k = 10$), spawn a fork of the environment. Observe that, even as BC approaches the terminus of its failing rollout, DARP continues to score highly, being precluded from doing so only about halfway through the Stack rollout and roughly 80% of the way through the Hopper rollout.

A.2.4 CAN DARP RECOVER FROM BC ERROR?

In order to interrogate DARP’s robustness to accumulated error, we roll out a BC agent in an environment in which it is known to fail, and, at every k steps, we spawn a fork of the environment and commence rolling out a DARP agent within that clone. The results, displayed in Fig. 12, reveal that, even as BC approaches failure and drifts away from the support of the expert demonstrations, DARP succeeds in recovering and scoring very highly. This finding suggests that DARP does indeed possess superior robustness to the accumulation of error, performing well in the marginally out-of-distribution states into which a failing BC agent drifts.

A.2.5 COMPARISON WITH CCIL

Environment	Method	Improvement over BC (%)
Hopper	CCIL	32.6%
	DARP	53.2%
Walker	CCIL	84.1%
	DARP	84.1%
Ant	CCIL	12.6%
	DARP	84.5%
HalfCheetah	CCIL	5.4%
	DARP	418.7%

Table 6: **Relative improvement over BC, juxtaposed with CCIL.** A comparison of DARP against CCIL on the standard MuJoCo benchmarks. DARP attains relative gains that equal or exceed those of CCIL across all tasks.

As reported in Table 6, we evaluate our method against CCIL, a baseline that induces smoothness explicitly. Drawing upon the scores reported in the CCIL paper, we compare the percentage improvement over BC. We observe that DARP outperforms CCIL substantially on three of the four environments, and by a particularly pronounced margin on HalfCheetah (418.7% versus 5.4% improvement).

A.2.6 DISTANCE METRIC SENSITIVITY

Although every experiment reported in this paper adopts Euclidean distance for the selection of nearby neighbors, it is natural to contemplate alternative metrics, such as cosine similarity, particularly within

Method	Success Rate (%)
BC	47
DARP (Cosine Similarity)	70
DARP (Euclidean Distance)	75

Table 7: **Distance Metric Sensitivity.** A comparison of success rates under R3M features. DARP with Euclidean distance and with cosine similarity perform comparably, surpassing BC by 28% and 23%, respectively. Results are reported on the Robosuite Stack task.

high-dimensional embeddings such as R3M. We find (as shown in Table 7) that DARP performs comparably—only 5% worse—when cosine similarity is substituted for Euclidean distance. This observation suggests that DARP is robust to the choice of distance metric employed.

A.2.7 DARP IN COMBINATION WITH DIFFUSION POLICY

Method	Stack	Threading	Peg Insertion
BC (MLP)	34	12	28
DARP (MLP)	62	38	40
Diffusion	52	44	54
DARP w/ Diffusion	82	72	66

Table 8: **DARP in combination with diffusion policy.** DARP confers substantial improvements when applied to both standard MLP policies and diffusion policies alike. Results are reported on the Robosuite Stack, Threading, and Peg Insertion tasks as success rates over 50 trials, under state-based observations. We note that the number of demonstrations employed here is considerably smaller than that used in Table 2, so that the baseline BC and DARP figures do not coincide.

Although all of our reported experiments are conducted with an MLP backbone, diffusion policy (Chi et al., 2024) has established itself as a state-of-the-art model class for imitation learning, particularly on manipulation tasks. Table 8 reveals that DARP is by no means mutually exclusive with diffusion; the two may be combined to yield an even more performant model. Deploying the DARP architecture atop a diffusion backbone outperforms DARP with an MLP backbone by success-rate increases of 20%, 34%, and 26% across the three tasks respectively, thereby surpassing standard BC by a cumulative 48%, 60%, and 38% on those same tasks.

Method	Training (s/epoch)	Inference (s/step)
Diffusion	5.610	0.15700
DARP (MLP)	0.559	0.00437

Table 9: **Computational efficiency of DARP with an MLP backbone relative to diffusion policy.** A comparison of the runtime costs incurred by DARP (MLP) and by diffusion policy. Diffusion policy proves markedly more expensive, being $\approx 10\times$ slower in training and $\approx 36\times$ slower at inference.

In addition, we find empirically that DARP with an MLP backbone is considerably faster than standard diffusion, most conspicuously at inference—see Table 9.

A.2.8 DARP TRAINED ON HUMAN DEMONSTRATIONS

The expert demonstrations used to train the models for the MuJoCo and Robosuite environments are collected by an optimal reinforcement learning policy. It is imperative to ascertain that DARP preserves its performance advantage over BC when trained instead on expert demonstrations furnished by humans. Indeed, when trained on human demonstrations for the Robosuite Stack task, DARP outperforms standard BC by 15% (see Table 10).

Method	Success Rate (%)
BC	45
DARP	60

Table 10: **Results with human demonstrations.** A comparison of success rates when training on human-provided data rather than on data collected by RL policies. Results are reported on the Robosuite Stack task.

A.2.9 CHOICE OF RETRIEVAL HYPERPARAMETERS

Environment	Method	Score (Mean)
Hopper	BC	507.49
	DARP	805.69
Stack	BC	0.12
	DARP	0.31

Table 11: **Performance comparison when validation loss governs the selection of training epochs and retrieval hyperparameters.** Mean scores on the Hopper and Stack environments. Observe that DARP preserves its performance advantage over BC.

Figure 9 reveals that certain selections of the retrieval parameters (a very low number of neighbors, for instance) cause DARP to perform worse than standard BC. We find, however, that choosing the retrieval hyperparameters so as to minimize validation loss constitutes an effective strategy for identifying performant settings; see Table 11.

A.3 PSEUDOCODE

We provide pseudocode for the DARP algorithm; see Algorithm 1.

A.4 RUNTIME ANALYSIS

Environment	k	Train (s/epoch)	Test (s/step)
Hopper (1,000 datapoints)	BC	0.102	0.00127
	100	0.126	0.00413
	200	0.128	0.00418
	300	0.130	0.00420
	400	0.131	0.00421
	500	0.130	0.00419
Stack (4,200 datapoints)	BC	0.431	0.00139
	100	0.556	0.00437
	250	0.539	0.00442
	500	0.559	0.00437
	750	0.576	0.00433
	1000	0.591	0.00478
	1500	0.682	0.00452
	2000	0.758	0.00451

Table 12: **Runtime comparison across environments.** Training and testing speeds (in seconds) for Behavior Cloning (BC) and for varying values of k on the Hopper and Stack datasets.

As reported in Table 12, we analyze the computational overhead of DARP at both training and inference time. Our analysis indicates that the computational cost scales sub-linearly with increasing

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Algorithm 1 Difference-Aware Retrieval Policies

```

1: Input: Expert demonstrations  $\mathcal{D}^* = \{(s_j^*, a_j^*)\}$ , number of neighbors  $k$ 
2: Initialize:  $f$  parameters  $\theta$ 
3: if  $g$  is parametric then
4:   Initialize:  $g$  parameters  $\psi$ 
5: end if
6: // Training Loop
7: while not converged do
8:   Sample batch of query data  $(s_q^*, a_q^*) \sim \mathcal{D}^*$ 
9:   for each query pair  $(s_q^*, a_q^*)$  in batch do
10:    // Find  $k$ -Nearest Neighbors from the entire dataset  $\mathcal{D}^*$ 
11:     $\mathcal{N}_k(s_q^*) \leftarrow \text{argmin-}k_j d(s_q^*, s_j^*)$ 
12:    // Compute Neighbor-based Predictions
13:    for each neighbor index  $i \in \mathcal{N}_k(s_q^*)$  do
14:       $a'_i \leftarrow f_\theta(s_i^*, a_i^*, s_i^* - s_q^*)$ 
15:    end for
16:    // Aggregate Predictions
17:    if  $g$  is parametric then
18:       $\hat{a}_q \leftarrow g_\psi(\{a'_i\}_{i \in \mathcal{N}_k(s_q^*)})$ 
19:    else
20:       $\hat{a}_q \leftarrow g(\{a'_i\}_{i \in \mathcal{N}_k(s_q^*)})$ 
21:    end if
22:  end for
23:  // Update Parameters based on the batch loss
24:   $\mathcal{L} \leftarrow \sum_{(s_q^*, a_q^*) \in \text{batch}} \|\hat{a}_q - a_q^*\|^2$ 
25:  // Gradient descent step
26:   $\theta \leftarrow \theta - \alpha \nabla_\theta \mathcal{L}$ 
27:  if  $g$  is parametric then
28:     $\psi \leftarrow \psi - \alpha \nabla_\psi \mathcal{L}$ 
29:  end if
30: end while
31: Output: Trained parameters  $\theta$  and, if applicable,  $\psi$ 

```

1350 k , and that the inference time remains tractable for real-time robotic application. On the Robosuite
1351 Stack task at $k = 500$, for instance, inference consumes approximately 0.00437 seconds per step on
1352 our hardware, which corresponds to a control frequency in excess of 230 Hz.
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